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SUPERSEDING
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MILITARY SPECIFICATION

MANUALS, TECHNICAL: REAL PROPERTY FACILITIES

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements for the development of manuals, used for maintenance and operation of real property facilities (see 6.3.1).

2. APPLICABLE DOCUMENTS

2.1 Unless otherwise specified the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38784 - Manual, Technical, General Style and Format Requirements.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. The manual(s) shall be furnished in the form(s) specified (see 6.2).

3.1.1 Manual outline. When manual outline is required (see 6.2), the outline shall be prepared as specified in 3.1.1.1.

3.1.1.1 The text plan shall be in accordance with the requirements of the applicable technical manual specification, and contain the number of paragraphs necessary to show the major elements of the equipment or system. Each paragraph title or notation shall be followed by a one-sentence statement outlining the proposed paragraph content.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.1.2 Technical accuracy. The manual shall accurately describe the system, equipment, or functions. Any technical errors or conflicting information will constitute reason for rejection of the manual.

3.1.3 Nomenclature, abbreviations, and symbols shall conform to MIL-M-38784.

3.2 Content. Manuals shall contain the following data, as applicable, and provide adequate instructions for installation, operation, maintenance, repair, and parts data for logistics support of the system or equipment.

Front matter
General information
Operation
Preventive maintenance
Trouble shooting
Corrective maintenance
Lubrication instructions
Drawings
Illustrations
Photographs
Diagrams
Notes, cautions, and warnings
Parts listings

The manual shall provide system description, purpose, capabilities, function, physical location within the facility, power distribution, signal distribution (electrical or otherwise), system operation accounting for modes of operation, system performance checks, and diagrams necessary for a full system treatment. Information included in subsystem or equipment manuals and pertinent to system analysis shall be included in order that all interface relationships are fully identified and described.

3.2.1 Front matter. Front matter shall normally be included as described in 3.2.1.1, 3.2.1.2, and 3.2.1.3. Succeeding volumes of a multivolume manual shall contain front matter as described in 3.2.1.1, 3.2.1.2, and 3.2.1.3.

3.2.1.1 Cover. The facility shall be identified on the cover by the name of the facility, name of the activity, and construction contract number. If the manual is in more than one volume, the volume number, the total number of volumes in the set, and the general content shall be included on each cover. Covers for multiple volumes for a single facility shall be similar. The security classification shall be included as specified (see 6.2).

3.2.1.2 Title page. The title page shall show the name of the preparing firm, the contract number, and the month and year of publishing. In addition, multivolume manuals shall show the general content on the title page.

3.2.1.3 Table of contents. The table of contents shall be consistent with the manual format and be in accordance with commercial practice.

3.2.2 General information. The manual shall include an overall description and purpose of the system or equipment. The functioning of the system or equipment and of its component parts shall be described. The description shall include the intended use, capabilities, and limitations of the system or equipment. If the manual covers more than one model of a system or equipment, or systems or equipment modified by field change, a description of the

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differences shall be provided. Quick reference data shall be included and shall describe technical or design characteristics of the equipment. Examples of such data are:

Descriptive (nameplate) data necessary to identify manufacturer, type, model, and the purpose of the facility, equipment, or subsystem which the manual covers.

Functional characteristics, such as:
- Power requirement
- Modes of operation
- Power output
- Frequency
- Pulse Characteristics
- Sensitivity: selectivity

Capabilities, such as:
- Rated ranges
- Speed
- Resolution
- Accuracy

Rated outputs, such as:
- Wattages
- Voltages
- Horsepower
- Gallons per minute
- Thrust
- Torque
- Footcandle or lumens
- Amperages
- Phase
- Power factor
- Lumens/Watt
- Rated life in hours
- British thermal units
- Ton refrigeration
- Cubic feet per minute
- Coefficient of performance
- Coefficient of utilization for lamps
- Interrupting capacity, RMS, AMP, asymmetrical

Special characteristics, such as:
- Operating temperatures
- Heat dissipation per unit
- Presure
- Humidity
- Tolerances
- Unusual materials or material applications.

3.2.3 Operation. Operating instructions shall include routine and emergency procedures (manual, automatic, local, and remote), and safety precautions. When the equipment can be turned on-off from an operating console, limits to be observed in the starting, operating, stopping, or shutting down of the equipment or system shall be provided. When the equipment turn-on is complicated, the operator shall be referred to the turn-on and checkout chart. Adequate illustrative material shall be provided to identify and locate operating controls and indicating devices. The function of each operating control and indicating device shall be included. Emergency operating instructions shall include alternate procedures to be followed when normal operation is not possible

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because of emergency conditions, such as power failure, control air failure, lubricating oil failure, or partial failure of requirements. Emergency operating instructions and procedures shall be located for quick and ready reference.

3.2.4 Preventive maintenance. Preventive maintenance information shall be provided. Use of special tools, materials, and test equipment shall be specified, including model/type designation, as appropriate. The following procedures shall be stressed, if applicable.

3.2.4.1 Cleaning and lubrication. Periodic cleaning and lubrication information, types of cleaning agents or lubricants required, recommended intervals, such as monthly, quarterly, semiannually, or hours of operation. Application points, type of lubricant and capacity (required amounts) shall be identified. Pictorial format for lubrication is desirable. Cleaning and lubrication required during repair, replacement, and reassembly shall also be covered (see 3.2.6).

3.2.4.2 Inspection. Instructions, including scheduling for inspection of equipment for damage and wear shall be included. Tabular or chart form is preferred and shall include, where applicable, allowable service limits, wear, backlash, end play, length and depth of scoring, and balance. These instructions shall be sufficiently complete to serve as standards by which experienced technicians may determine when parts may be continued in use and when they must be replaced.

3.2.4.3 Performance and adjustment verification. Instructions shall be included for verification of the accuracy of equipment measurement devices needed to restore the equipment to its original state. These instructions shall provide complete step-by-step procedures which will enable the user to check the accuracy of the indications or readings. The location of test connections and the values expected at these points shall be included, preferably in illustrated form. Adjustment of built-in self-test features shall be included. Data shall include a list of the additional equipment required to accomplish the verification, such as temperature, vacuum, pressure, hydraulic, or pneumatic gages; voltmeters; ammeters, frequency meters; or signal generators. The listing shall show methods of use or application, range of scales, and specific minimum tolerances or percentages of accuracy. This information is preferred in tabular format. Such headings as the following may be used:

NOMENCLATURE	PART/MODEL NO.	APPLICATION	RANGE	ACCURACY

3.2.4.4 Exercising emergency standby equipment. The frequency and special procedures for exercising emergency standby equipment shall be included, as applicable.

3.2.5 Troubleshooting. Failures that might occur during operation of equipment shall be listed. Troubleshooting data and fault isolation techniques shall state: (a) the indication or symptom of trouble, (b) the sequential instructions necessary, including test hookups, to determine the cause, (c) special tools and test equipment, and (d) methods for returning the equipment to operating conditions. Information may be in chart form or in tabular format with appropriate headings.

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3.2.6 Corrective maintenance. Instructions shall be included, but not limited to, all removal, repair adjustment, and replacement procedures. Exploded and sectional views giving details of assemblies shall be provided, as necessary, to clarify the test. For mechanical items, dimensional information with tolerances, clearances, wear limits, maximum bolt-down torques, and in-place balancing or other means of reducing noise level, if required, shall be supplied. Requirements for cleaning and lubrication prior to reassembly shall be included, if critical. Information on the use of special tools and test equipment supplied with the equipment, as well as any cautions or warnings which must be observed to protect personnel and equipment, shall also be included (see 3.2.4.1).

3.2.7 Notes, cautions, and warnings. Notes, cautions, and warnings shall be used to emphasize important and critical instructions where necessary. Notes, cautions, and warnings shall immediately precede the applicable instructions and shall be selected as follows:

NOTE - concerns an operating procedure or condition which should be highlighted.

CAUTION - concerns an operating procedure or practice, which if not strictly observed, could result in damage to, or destruction of equipment.

WARNING - concerns an operating procedure or practice, which if not strictly observed, could result in a hazard to health, injury to personnel, or loss of life.

3.2.8 Parts lists. Parts lists shall provide positive identification of parts necessary for support of the systems or equipment and shall include sufficient information to enable maintenance personnel to requisition replacement parts.

3.2.8.1 Systems or top assembly lists. The parts list for a system shall identify all parts necessary to accomplish system operations that are not already included (listed) with subsystem or equipment lists. Such parts to be listed would include, but not be limited to, such items as cable, pipe, flanges, feed-through devices for cables, connection or junction boxes, connection devices, and distribution panels.

3.2.8.2 Parts list illustrations. Clear and legible illustrations are to identify component parts and parts relationships. Part numbers and part names may be shown on illustrations or separately listed. When the illustrations omit the part numbers and part names, both the illustrations and separate listings shall cross-reference illustrated part to listed part. Parts in the listing shall be grouped by systems, subsystems, and components.

3.2.9 Pollution control. All pertinent pollution control information shall be included. The information may be in the form of operating instructions, maintenance instructions, or both, for prevention or correction of pollution.

3.3 Format.

3.3.1 Use of existing manuals. Wherever possible, commercial manuals will be incorporated without change in either content or format. The commercial

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manuals may be bound without disassembly in the facility manual, or may be disassembled and appropriate portions incorporated. (When manufacturer's manuals contain several models, any information not applicable to the specific model installed shall be edited from the project manual).

3.3.2 General arrangement of data. The data may be grouped by system, function, or location within the facility, and the order of arrangement shall be consistent throughout the manual. Final determination of the arrangement of data shall be subject to approval of the procuring activity.

3.3.3 Volumes. The manual may be divided into volumes to conform to the general arrangement of data, or to prevent the manual from becoming unwieldly in size.

3.3.4 Text. The text shall be specific, concise, and clearly worded to be easily understood by personnel involved in the operation, maintenance, and repair of the equipment. The manual shall be oriented toward operation, maintenance, and repair of the equipment by the operators and maintenance and repair trades without the assistance of a manufacturer's representative. Descriptive matter and theory shall be limited to those technical details which are essential to a proper understanding of the system or equipment for satisfactory operation, maintenance, and repair.

3.3.5 Illustrations. Manuals shall contain illustrations, drawings, or photographs necessary to locate and identify components of operational and maintenance significance. Where necessary for clarity, illustrations shall show configuration and the removal and disassembly of parts. The following shall be included: schematic diagrams which show the arrangement of component devices or parts; wiring diagrams which show the connections of the circuit arrangement; and schematic piping diagrams which show the interconnection of components, of piping, tubing, or hose, and the direction and sequence of fluid flow, drawings, and photographs.

3.3.5.1 Circuit diagrams. Circuit diagrams for electronic units shall be included to support maintenance and troubleshooting. Circuit diagrams shall cross-reference repair parts shown in test tables and parts lists. The function name of each stage or circuit, primary signal flow, test points, wave forms with pertinent characteristics, electrical characteristics of parts name of each variable control, wire size and color code identification, input and output connectors/terminals voltages and signals shall be specified. Voltage and current resistance values measured with controls set for normal operation shall be shown for significant points, such as terminal boards, tube sockets, and connectors. The direct current resistance of coils, chokes, and transformers shall be shown at the part (one ohm or more only). Interconnecting cable diagrams shall be furnished to show TO-FROM information, including any intermediate connections. Block diagrams shall be provided to support installation instructions, but shall not be substituted for necessary schematic diagrams. Logic diagrams for nonrepairable digital circuitry may be substituted for schematic diagrams; however, schematic diagrams for power supplies, relays, transformers, variable controls, tube (including cathode-ray tube (CRT), and photocells shall be included.

3.3.6 Warning page. A warning page, consisting of the more vital warnings extracted from those shown throughout the manual, shall be used when care must be exercised in areas of potential danger, such as high voltage, radioactive

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material, toxic chemicals, flammable liquids, explosive materials, carcinogens, or high pressures. The warning page shall be placed on the inside cover or in front of the initial page(s) of the manual.

3.3.7 Manuscript copies for review. When specified, manuscript (draft) copies provided for Government review shall be in the format and number as specified (see 6.2).

3.3.8 Reproducible copy. When specified (see 6.2), a reproducible copy shall be furnished and shall conform to MIL-M-38784.

4. QUALITY ASSURANCE PROVISIONS

4.1 Manuscript reviews. Unless otherwise specified (see 6.2), a manuscript copy shall be submitted with each submittal of the plans and specifications for the facility. The manuscript submitted shall be at the same phase of development as the plans and specifications submitted.

4.2 Validation. Operating and maintenance procedures, including checkout, calibration, alinement, scheduled removal and replacement instructions, and associated check lists shall be validated against the system (or equipment) in the presence of Government personnel, as specified (see 6.2 and 6.3.4).

5. PACKAGING

5.1 Packaging. Unless otherwise specified in the contract, the material furnished in accordance with this specification shall be packaged as specified herein.

5.1.1 Manuscript copy. The manuscript copy shall be packaged flat in containers or wrapped separately. The test material and artwork shall be packaged together in one package or in separate packages. Copies of large artwork to be used for review only, may be folded. The packaged manuscript and artwork shall be packed in suitable exterior containers that will insure arrival at destination in a satisfactory condition and will be acceptable to the carrier.

5.1.2 Reproducible copy and artwork. The reproducible copy and artwork shall be packaged flat. Unless otherwise directed, artwork shall not be folded or rolled. The interior packaging material shall be waterproof and shall not contain chemical substance that would damage the reproducible copy or artwork. The packaged reproducible copy and artwork shall be packed in suitable exterior containers as specified in 5.1.1 above.

5.2 Marking. In addition to the sender and addressee markings, the exterior of each package shall be marked with the following information. The marking shall be applied with waterproof material and shall be legible.

a. Contract or purchase order number.
b. Publication identification number.
c. The word(s) MANUSCRIPT, REPRODUCIBLE COPY OR ARTWORKS, as applicable.
d. Quantity of copies in interior and exterior packages.
e. Others (specify).

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6. NOTES

6.1 Intended use. The manuals are intended for use in operations and maintenance of complete real property facilities, including such components as roofing, special flooring, piping, valves, and structural items in addition to installed equipment, such as air handling equipment, standby electrical power generators, or sound systems.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. The form(s) in which the manual(s) is to be furnished (see 3.1).
c. Whether a manual outline is required (see 3.1.1).
d. When security classification is required (see 3.2.1.1).
e. The format and number of manuscript copies to be submitted for review (see 3.3.7).
f. Whether reproducible copy is required (see 3.3.8).
g. When manuscript copy is not to be submitted (see 4.1).
h. Validation procedures required (see 4.2 and 6.3.4).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No.	Data requirements title	Applicable DID No.	Option
3.1	Manual, Technical Standard	DI-M-2044A	

(Data Item Descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, AMSDL. Copies of Data Item Descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Definitions.

6.3.1 Real property facilities. Buildings, structures, utilities systems, improvements, and appurtenances thereto. Includes equipment attached to and made part of buildings and structures, such as heating systems, but not movable equipment, such as plant equipment.

6.3.2 Commercial manuals. A manufacturer's manual, instructions, or other data containing technical information concerning assembly, installation, operation, maintenance, and parts identification, in card, looseleaf, or bound manual form, which is normally furnished by a manufacturer to purchasers of his commercial products or sold to repair or service men or repair shops.

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6.3.3 Procuring activity. As used in this specification, the procuring activity is the Government organization which is assigned review and approval authority by the contract under which the facility manual is prepared.

6.3.4 Validation. Validation is the process by which the contractor proves a manual for technical accuracy and adequacy. This is accomplished by actual performance of the maintenance and operation instructions on the system (or equipment) for which the manual was written. Validation entails the actual performance by contractor personnel of operating and maintenance procedures, including checkout, calibration, alinement, scheduled removal and replacement instructions and associated check lists, to insure technical accuracy and adequacy. The extent of validation will be determined by the magnitude, complexity, and unusuality of the procedure being proved. Specific validation requirements shall be specified in the contract.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:

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Preparing activity:

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Project No. TMSS-N119

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